

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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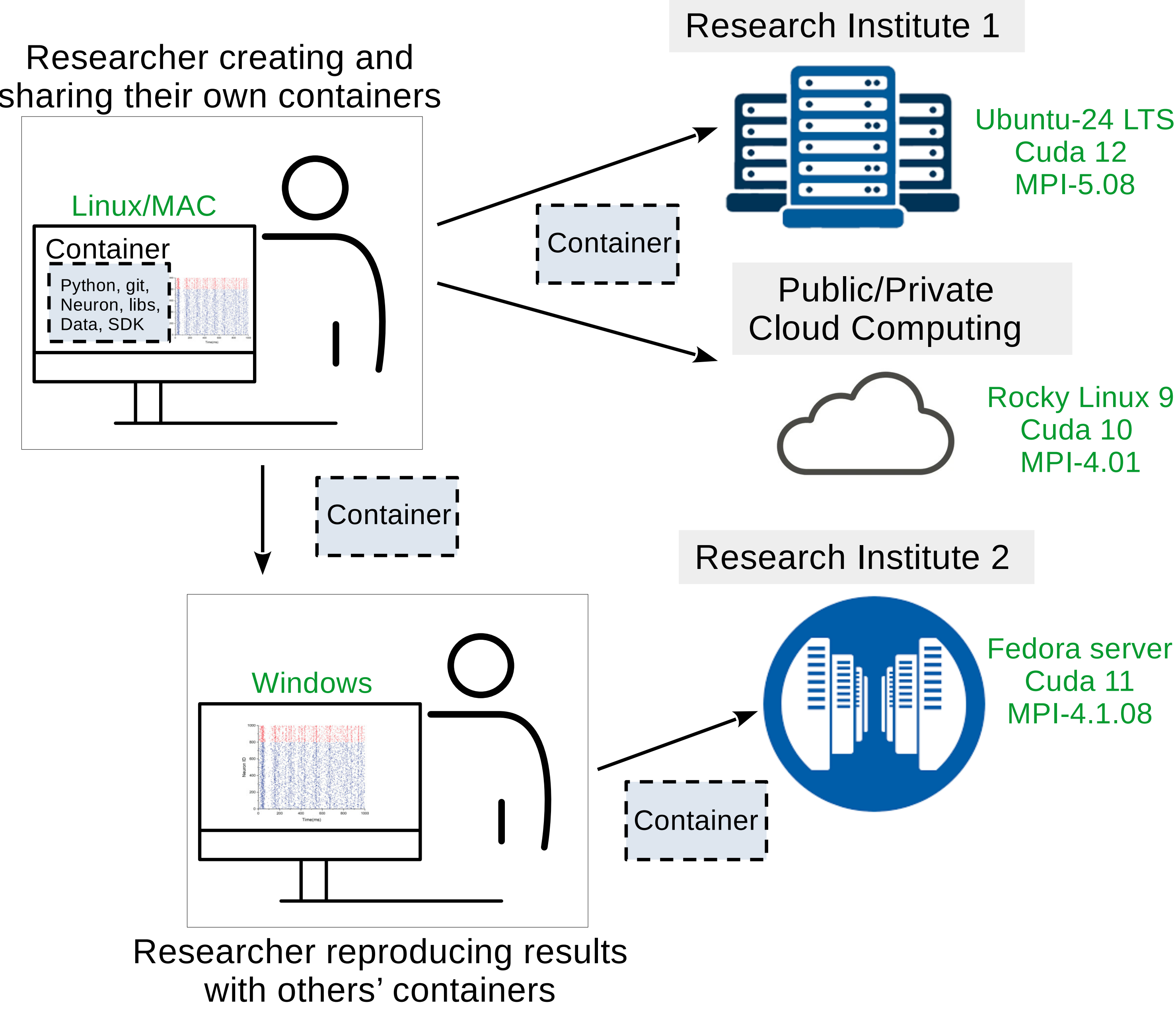
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Introduction

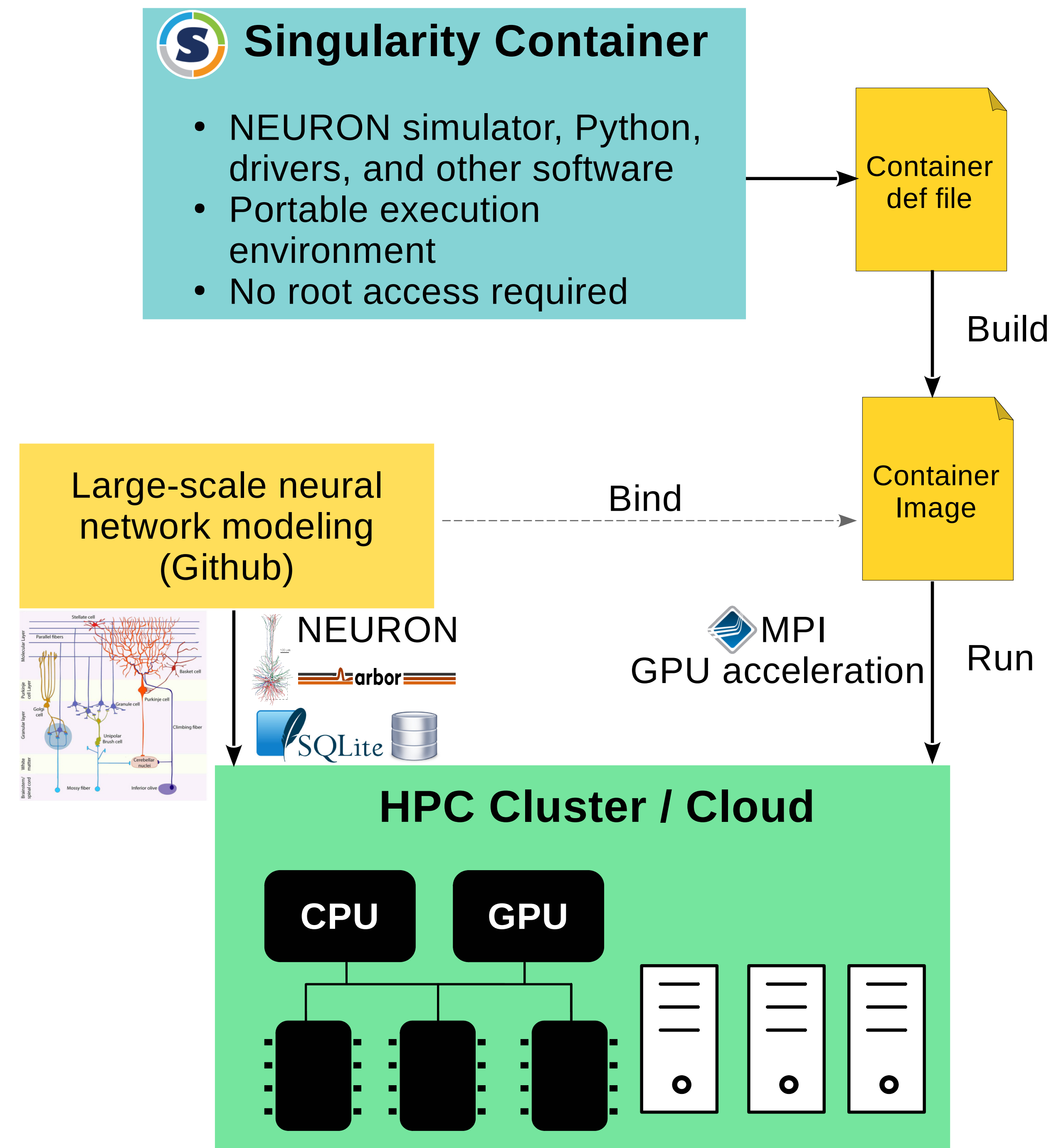
1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.
3. We address this issue using **Singularity containers** that encapsulates the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber–granule cell network** simulations.

Containerization for Scientific Computing

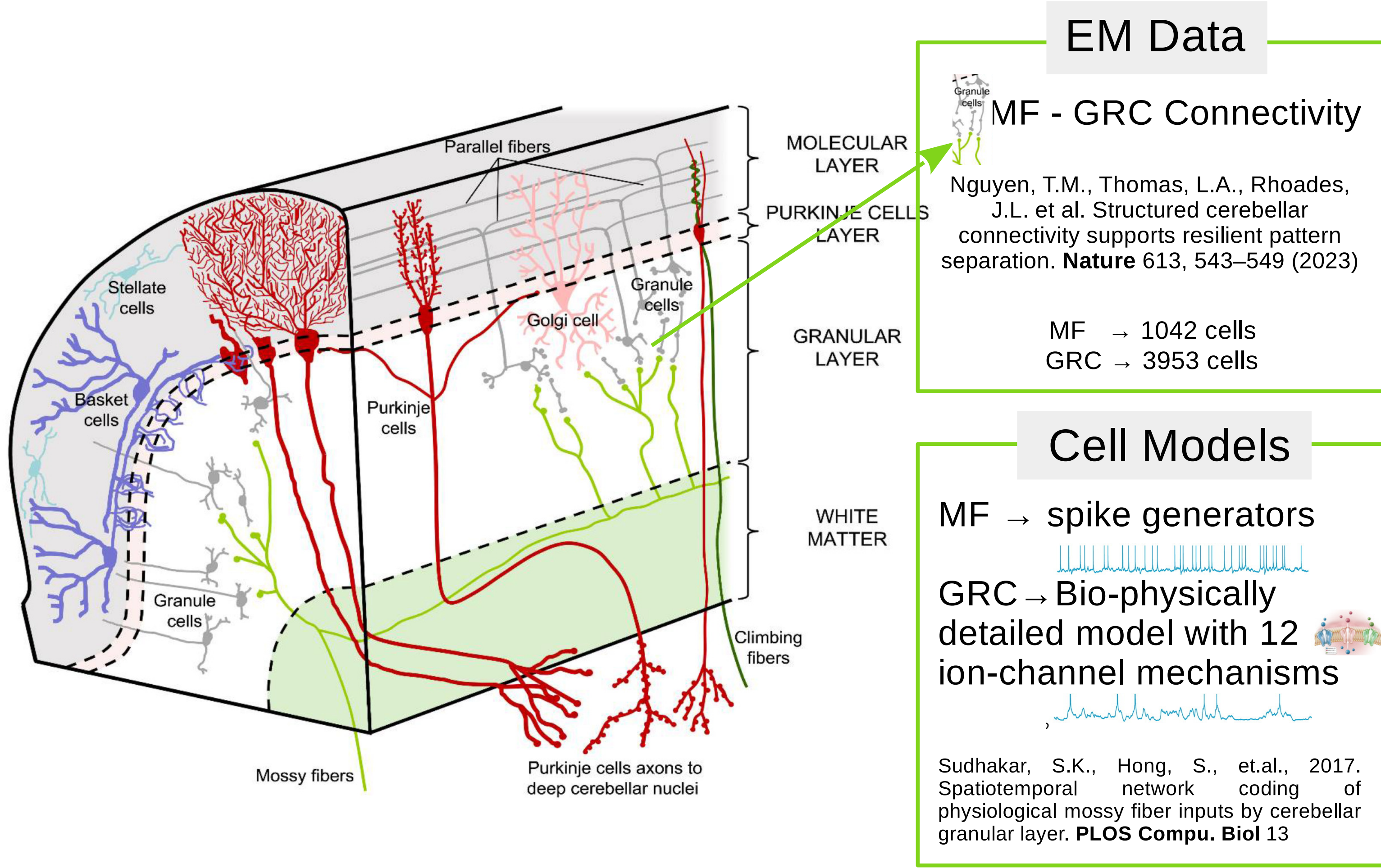
A container is a lightweight, standalone, virtualized executable environment that includes all dependencies required to run large-scale scientific models.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple copies ($n = 80$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

Table: Network Configuration and Simulation Parameters

Number of MF	85,600
Number of GRC	314,000
Number of compartments	942,000
Number of synapses	990,960
Simulation time	2 s
Simulation software	CORENeuron

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU

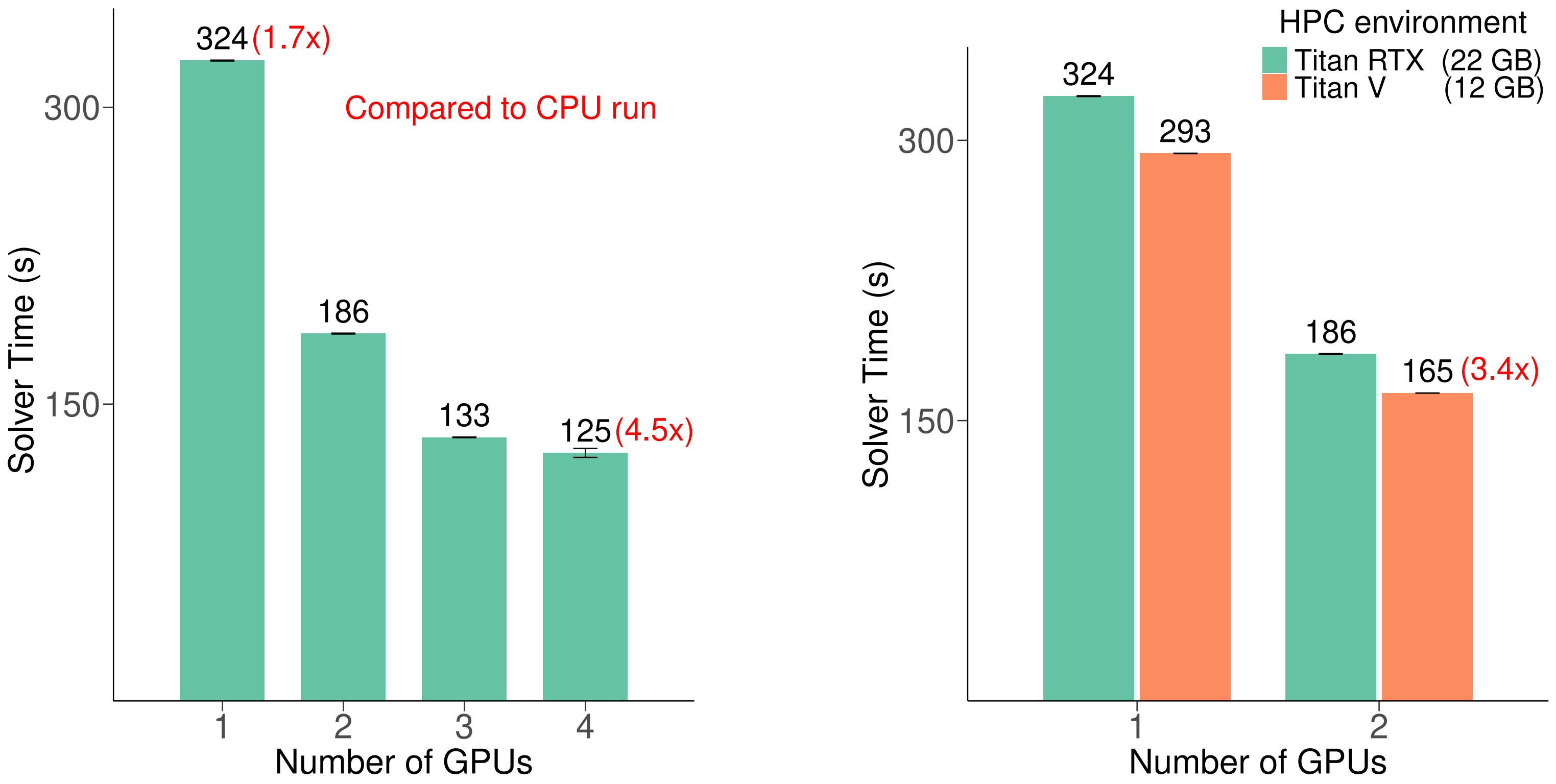


Figure: Solver time comparisons in CORENeuron simulations across HPC platforms.

TITAN RTXs outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

- 🔧 **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.
- 🔧 **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

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